1/1/2025.

NUMERICAL AND STATISTICAL TECHNIQUES

Analytic methods have very limitation. Numerical methods are many iterative. so a system can be trained to get solution.

example: Analytic Method

$x-1=0$

$\Rightarrow x=1$.

Numerical method:

$f(x)=x-1$ let $x_0=0$, $x_{n+1}=x_n+0.5$

$f(x_0)=0-1\neq 0$.

$f(x_1)=0.5-1\neq 0$.

$f(x_2)=-0.5+0.5=0$, $1-1=0$

$\therefore x_2=1$ is the solution

Error analysis

Error is the difference between - True number [exact solution] and approximate number [approximate]

$E = x_T - x_A$.

Absolute error

$E_A = |x_T - x_A|$

Relative error

$E_R = \frac{|x_T - x_A|}{|x_T|}$.

Percentage error

$E_P = E_R \times 100$.

Inherent error

Error is a problem due to available approximate data and available computing aids. & can be minimized using better data and by using computing aid with high precision.

Round-off error

$E = x_T - x_A$ let $x_T = 1.3415632$; $x_A = 1.3416$ [Symmetric rounding]

chopping error

$x_T = 1.34562718$; $x_A = 1.34562$ [$E = x_T - x_A$]

Truncation error

$E = x_T - x_A$. $x_T = 2.433333\ldots$; $x_A = 2.433$. [$E = x_T - x_A$]

Q1] Find the error in the approximation of $e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \frac{x^6}{6!}$ when $x = 1/5$.

case 1 : Taking first 3 terms

case 2 : Taking first 4 terms

b) $E = x_T - x_A$

for case I

$E = \left[\frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \frac{x^6}{6!}\right]_{x=0.2}$

case II.

$E = \frac{x^4}{4!} + \frac{x^5}{5!} + \frac{x^6}{6!}$

$= 6.942222 \times 10^{-5}$

2. Find maximum absolute error, relative error and percentage error of $u = \frac{x^3y^3}{z}$, where $x = y = z = 0.1$ and $\Delta x = \Delta y = \Delta z = 0.002$.

$$\Delta u = \left|\frac{\partial u}{\partial x}\right| \Delta x + \left|\frac{\partial u}{\partial y}\right| \Delta y + \left|\frac{\partial u}{\partial z}\right| \Delta z.$$

$$\left|\frac{3x^2y^3}{z}\right|_{x=0.1} 0.002 + \left|\frac{3y^2x^3}{z}\right|_{0.1} \times 0.002 + \left|\frac{x^3y^3}{z^2} \times 0.002\right|$$

$\Delta u = 1.4 \times 10^{-6}$.

relative error $= \frac{\Delta u}{u} = \frac{1.4 \times 10^{-6}}{1 \times 10^{-5}} = \frac{1}{10} = 0.1$ or 0.14.

Percentage error = 14%.

Find ε_A if $x_A = 67.84$ and $\varepsilon_R = 1\%$.

$\varepsilon_R = \frac{\varepsilon_A}{x_T}$ or $\frac{\varepsilon_A}{x_A}$ $\Rightarrow \frac{67.84}{100} = \varepsilon_A \Rightarrow \varepsilon$

Numerical solution of algebric and Transcendental.

Algebric:
algebric equation is $f(x) = 0$ where $f(x)$ is contain polynomial of in x

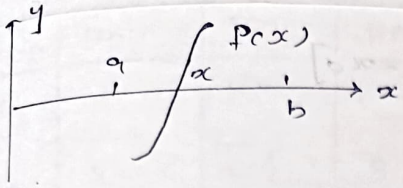
Transcendental equation:
$f(x) = 0$ where $f(x)$ contains powers of x, logarithmic, exponential terms.

Methods of solution:

1) Bisection Method [Bolzano]
2) Newton Raphson method
3) Secant method.
4) Successive iteration method. [Fixed point iteration]

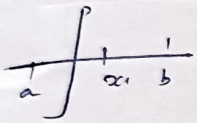
1) Bisection method.

$f(x) = 0$. $\Rightarrow x$ is the solution $\Rightarrow$ $f(x)$ touches x axis at x.

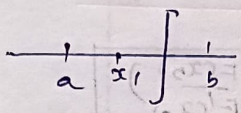


* find a and b such that $f(a) < 0$ and $f(b) > 0$.
the root x lies in between a and b.
$x \in [a, b]$.

* $a_1 = \frac{a+b}{2}$.



$f(x_1) > 0$
$x_1 < b$
$x \in [a, x_1]$
$a = a$
$b = x_1$
$x_2 = \frac{a+x_1}{2}$

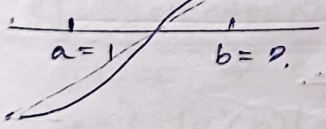


$f(x_1) < 0$ $x_1 > a$
$x \in [a, b]$
$a = a$ $b = b$
$x_2 = \frac{b+x_1}{2}$

and so on.

1] find the root of $x^3 - x^2 - 1 = 0$ using Bisection method

$f(x) = x^3 - x^2 - 1$.



$f(0) = -1$
$f(1) = -1 < 0$
$f(2) = 3 > 0$
$a = 1$ $b = 2$.

It no.	a	b	f(a)	f(b)	$x_0 = \frac{a+b}{2}$	f(xn)
1	1	2.	-1	3	1.5	0.125
2	1	1.5	-1	0.125	1.25	-0.6093375
3	1.25	1.5	-0.609 3375	0.125	1.3875	-0.2 91056
4	1.3375	1.5	-0.29101 56	0.125	1.43375	-0.095 94
5	1.43375	1.5	-0.095 94	0.125	1.468875	+0.0 111999
6	1.43375	1.468875	-0.095	0.0111 999	1.41531	-0.0 04327

(7)

a	b	f(a)	f(b)	xn
1.41531	1.468875	-0.04327	0.0111 999	1.460 92

(8)

a	b	f(a)	f(b)	xn
1.416092	1.468875	-0.016024	0.0111 999.	1.2164 835

$F(x_n) = -0.016$ 264

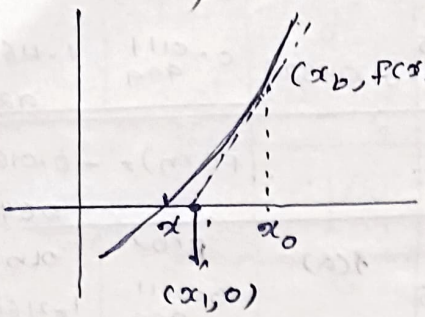
$f(x_n) = 0.016$ 264

since $x_{n-1} = x_8$ upto 2 decimal.

$x = 1.46$.

$\therefore$ root of equation = 1.46.

2) Newton Raphson Method



$$(y_1 - y_0) = m[x_1 - x_0]$$

$$= F'(x_0)$$

$$[0 - F(x_0)] = F'(x_0)[x_1 - x_0]$$

$$x_1 - x_0 = \frac{-F(x_0)}{F'(x_0)}$$

$$x_1 = x_0 - \frac{F(x_0)}{F'(x_0)}$$

Generally: $x_{n+1} = x_n - \frac{F(x_n)}{F'(x_n)}$

After a stage $x_n = x_{n+1}$ upto required decimal places then iteration can be stopped then $\alpha = x_n = x_{n+1}$ is the root

1] Find the root of equation $x^3 - x^2 - 1 = 0$ correct 3 decimal using Newton's method

$F(x) = x^3 - x^2 - 1$

$F'(x) = 3x^2 - 2x$

It No.	x_n	$F(x_n)$	$F'(x_n)$	$x_{n+1} = x_n - \frac{F(x_n)}{F'(x_n)}$
1	1	-1	1	$x_1 = 1 - \frac{-1}{1}$ $= 2$
2	2	3	8	$x_2 = 2 - \frac{3}{8} = 1.625$
3	1.625	0.650390	4.6718875	$x_3 = 1.625 - \frac{0.65039}{4.6718875} = 1.485786$
4	1.485786	0.0712379	3.651066	$x_4 = 1.485786 - \frac{0.0712379}{3.651066} = 1.465955$
6	1.465955	0.0013485	3.515162 1	$x_5 = 1.4651$

Since $x_4 = x_5 = 1.465$ correct to 3 decimal

It is the root $x = 1.465$

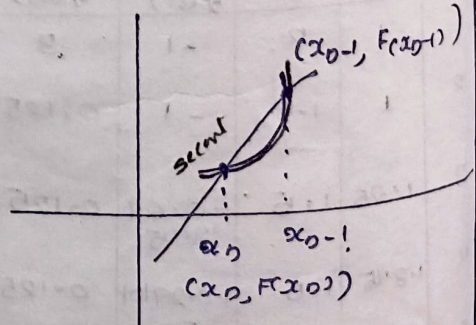
3) Secant method

By newton-raphson method $x_{n+1} = x_n - \frac{F(x_n)}{F'(x_n)}$ — (1)

Substituting (2) in (1)

$$x_{n+1} = x_n - \frac{F(x_n)}{F'(x_n)}$$

$$x_{n+1} = x_n - \frac{(x_n - x_{n-1}) \times F(x_n)}{F(x_n) - F(x_{n-1})}$$



$$F'(x_n) = \text{Slope } m = \frac{F(x_n) - F(x_{n-1})}{x_n - x_{n-1}}$$ — (2)

? find a root of the equation using secant method correct to 3 decimal $x^3 - x^2 - 1 = 0$.

$f(x) = x^3 - x^2 - 1 = 0$; $x_0 = 1$, $x_1 = 2$,

#No	x_{n-1}	x_n	$F(x_{n-1})$	$F(x_n)$	$x_{n+1} = x_n - \frac{x_n - x_{n-1}}{F(x_n) - F(x_{n-1})} \times F(x_n)$.
1	1	2.	-1	3	$x_2 = 2 - \frac{2-1}{3+1} \times 3 = 2 - \frac{3}{4} = 1.25$
2	2	1.25	3	-0.60937	$x_3 = 1.25 + \left[\frac{1.25-2}{-3-0.60937}\right] \times 0.60937$ $= 1.376629$.
3	2 1.25	1.376622	-0.60937	-0.2862	$x_4 = 1.376629 - \left[\frac{1.376629 - 1.25}{-0.2862 - 0.60937}\right] \times$ $= 1.488758$
4	1.376622	1.488 758.	-0.2862.	0.083283	$x_5 = 1.48875 - \frac{1.488758 - 1.376629}{0.083283 + 0.2862} \times$ $= 1.463507$
5	1.488875	1.46350	0.083283	-0.00723	$x_6 = 1.46350 + \left[\frac{1.46350 - 1.48875}{-0.00723 - 0.083}\right] \times 0.00723$ $= 1.465$
6	1.46350	1.4655	-0.00723.	-0.000025	$x_7 = 1.4655 + \left[\frac{1.4655 - 1.46350}{-0.000025 + 0.00}\right] \times 0.000025$ $= 1.4655$

since $x_6 = x_5 = 1.465$ correct upto 3 decimal 1.465 is the root

4] Successive Iteration Method / Fixed point Iteration

$f(x) = 0 \Rightarrow x = g(x)$. find 2 numbers a and b such that $|g'(a)| < 1$ and $|g'(b)| < 1$, to test convergence. then from general equation $x_{n+1} = g(x_n)$, start iteration with $x_0 = a$ and $x_0 = b$.

After a stage $x_{n+1} = x_n$. correct to some decimal values. which is the root.

7 find $x^3 - x^2 - 1 = 0$ correct to 3 decimal places using successive iteration.

$f(x) = x^3 - x^2 - 1$

$x^3 = 1 + x^2$

$x = [1 + x^2]^{1/3}$.

$x = g(x) = [1 + x^2]^{1/3}$

$g'(x) = \frac{1}{3}[1 + x^2]^{-2/3} \times 2x$

$g'(1) = \left| \frac{1}{3} \times [2]^{-2/3} \times 2 \right|$

$= \frac{1}{3} \times \frac{2}{(4)^{1/3}}$

$= 0.419973$.

$g'(2) = 0.4554$

	x_n	$x_{n+1} = (1 + x_n^2)^{1/3}$.
1	1	$x_1 = (2)^{1/3} = 1.259921$
	1.259921	$x_2 = [1 + (1.259921)^2]^{1/3} = 1.372844$
	1.372844	$x_3 = (1 + (1.372844)^2)^{1/3} = 1.42936$.
	1.42936	$x_4 = (1 + (1.42936)^2)^{1/3} = 1.446896$
	1.446396	$x_5 = [1 + 1.446396^2]^{1/3} = 1.456854$.
	1.456854	$x_6 = [1 + 1.456854^2]^{1/3} = 1.461606$
	1.461606	$x_7 = [1 + 1.461606^2]^{1/3} = 1.463767$
	1.463767	$x_8 = [1 + 1.463767^2]^{1/3} = 1.464750$.
	1.4647	$x_9 = 1 + 1.4647504^2]^{1/3} = 1.46519$
	1.46519	$x_{10} = [1 + 1.46519^2]^{1/3} = 1.4653$

since $x_9 = x_{10} = 1.465$ correct upto 3 decimal $x = 1.465$ is the root.

NUMERICAL METHOD TO SOLVE SYSTEM OF EQUATIONS IN LINEAR ALGEBRA

7/1/2025.

GAUSS SEIDEL ITERATION METHOD

[1] To solve $a_{11}x + a_{12}y + a_{13}z = b_1$
$a_{21}x + a_{22}y + a_{23}z = b_2$
$a_{31}x + a_{32}y + a_{33}z = b_3$.

Arrange the system as diagonally dominant.

$|a_{ii}| \geq \sum_{i \neq j} |a_{ij}|$

[2] find $x = \frac{1}{a_{11}}[b_1 - a_{12}y - a_{13}z]$

$y = \frac{1}{a_{22}}[b_2 - a_{21}x - a_{23}z]$, $z = \frac{1}{a_{33}}[b_3 - a_{31}x - a_{32}y]$

[If diagonally dominant]

[3] start iteration with initial value: $x^{(0)} = y^{(0)} = z^{(0)} = 0$

$x^{(1)} = \frac{1}{a_{11}}[b_1 - a_{12}y^{(0)} - a_{13}z^{(0)}]$ most recent iterated value

$y^{(1)} = \frac{1}{a_{22}}[b_2 - a_{21}x^{(1)} - a_{23}z^{(0)}]$

$z^{(1)} = \frac{1}{a_{33}}[b_3 - a_{31}x^{(1)} - a_{32}y^{(1)}]$

continue to find $x^{(n)}$, $y^{(n)}$, $z^{(n)}$ until $x^{(n+1)} = x^{(n)}$, $y^{(n+1)} = y^{(n)}$, $z^{(n+1)} = z^{(n)}$ correct to above decimal value

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solve ~~$2x+4y+6z=72$~~ $6x+15y+2z=72$, $27x+6y-z=85$, $x+y+54z=110$ using ~~write~~ gauss seidel method

* checking diagonal dominancy: $|6| \not\geq |15|+|2|$
∴ re arranging the sly.

$27x + 6y - z = 85$
$6x + 15y + 2z = 72$
$x + y + 54z = 110$

This system satisfies the condition ie it is diagonally dominant.

$x^{(0)} = y^{(0)} = z^{(0)} = 0$

$x^{(1)} = \frac{1}{27}[85 - 6y + z]$.

$y = \frac{1}{15}[72 - 6x - 2z]$

$z = \frac{1}{54}[110 - x - y]$.

$x^{(1)} = .3.141481418$

$y^{(1)} = .\frac{1}{15}[72 - 6[3.141481418]]$

$= 3.540740$

$z^{(1)} = \frac{1}{54}[110 - x^{(1)} - y^{(1)}]$

$= 1.913168741$

$x^{(2)} = \frac{1}{27}\left[85 - 6^{(y)} + z^{(1)}\right] = 2.432175$

$y^{(2)} = \frac{1}{15}\left[72 - 6^{(x)} - 2(8^{(1)})\right] = 3.572070$

$8^{(2)} = \frac{1}{54}\left[110 - x^{(1)} - y^{(1)}\right] = 1.925847.$

$x^{(3)} = \frac{1}{27}\left[85 - 6\left[3.572070\right] + 1.925847\right] = 2.425682$

$y^{(3)} = \frac{1}{15}\left[72 - 6\left[2.425682\right] - 2\left[1.925847\right]\right] = 3.572947$

$z^{(3)} = \frac{1}{54}\left[110 - 2.425682 - 3.572947\right] = 1.925951.$

$x^{(4)} = \frac{1}{27}\left[85 - 6\left[3.572947\right] + 1.925595\right] = 2.425491$

$y^{(4)} = \frac{1}{15}\left[72 - 6\left[2.425491\right] - 2\left[1.925595\right]\right] = 3.573010$

$z^{(4)} = \frac{1}{54}\left[110 - 2.425491 - 3.573010\right] = 1.925953.$

$x^{(5)} = \frac{1}{27}\left[85 - 6\left[3.573010\right] + 1.925953\right] = 2.42547$

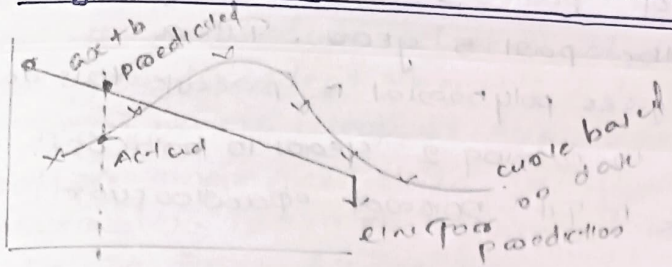
$y^{(5)} = \frac{1}{15}\left[72 - 6\left[2.425491\right] - 2\left[1.925951\right]\right] = 3.573018$

$z^{(5)} = \frac{1}{54}\left[110 - 2.42547 - 3.573018\right] = 1.925596$

since $x^{(4)} = x^{(5)} = 2.425$
$y^{(4)} = y^{(5)} = 3.573$
$z^{(4)} = z^{(5)} = 1.925.$

} these are the solution.

Polynomial Regression for Machine learning - least square method



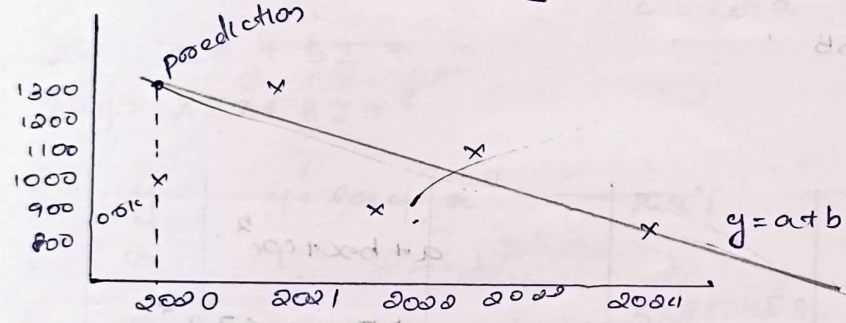
To minimize the diff b/w actual and real data

Suppose we have a set of data to forecast the future data using a computer. we shall create regression model The o/p can be displayed using programming language like python. To create more accuracy in python, prediction's polynomials of higher degree can be used.

example: Data

year	2020	2021	2022	2023	2024
No. of patients	1000	1300	900	1100	800

when we plot this data in graph we will get scattered diagram



we try to find a curve that can be a general representation [fits to all data]. for this purpose we need to minimize distance between real data and predicted data. Suppose $y = a+bx$ be the curve which passes through (x_1, y_1) (x_2, y_2) (x_3, y_3) we need to minimize the distance between predicted and original $|(y_1 - a+bx_1)|$

$\therefore$ we consider the function $f = \sum_{j=1}^{n} [y_j - a - bx_j]^2$.

using the methods in calculus find minima.

consider the necessary condition: $\frac{\partial f}{\partial a} = 0$ and $\frac{\partial f}{\partial b} = 0$

$\partial f/\partial a$: $-2 \Sigma (y_j - a - bx_j) = 0$.

$\partial f/\partial b$: $-2 \Sigma [y_1 - a - bx_j] x_j = 0$.

$\Rightarrow$

$$\Sigma y_i = \Sigma a + b \Sigma x_j$$
$$\Sigma x_j y_j = a \Sigma x_j + b \Sigma x_j^2$$

Similarly the normal equation for 2nd degree polynomial

$y = a+bx+cx^2$.

$\Sigma y = \Sigma a + b\Sigma x + c\Sigma x^2$.

$\Sigma xy = a\Sigma x + b\Sigma x^2 + c\Sigma x^3$

$\Sigma x^2 y = a\Sigma x^2 + b\Sigma x^3 + c\Sigma x^4$.

using calculator we can solve this system and find the unknown's to work general form and forecasting the data.

? The following data gives no. of patients affected by fever in the month of July for the past 5 years. Fit a first degree and second degree polynomial to forecast this data and predict the no. of patient in coming 2 years in both cases.

let $y = ax + b$ be the straight line to fit normal equation are

$$\Sigma y = na + b\Sigma x$$

$$\Sigma xy = a\Sigma x + b\Sigma x^2$$

year x	no. of patient y	xy	x^2
1	1000	1000	1
2	1300	2600	4
3	900	2700	9
4	1100	4400	16
5	800	4000	25
15	5100	14,700	55

$$5100 = 5a + 15b.$$

$$14,700 = 15a + 55b.$$

$a = \bar{x} = 1200.$

$b = \bar{y} = -60$

$[y = 1200 - 60x]$

$y = 840.$ at $x = 6$

$y = 780$ at $x = 7$

To fit $y = a + bx + cx^2$

$a + bx + cx^2.$

x	1	2	3	4	5
y	1000	1300	900	1100	800

$$\Sigma y = na + b\Sigma x + c\Sigma x^2$$

$$\Sigma xy = a\Sigma x + b\Sigma x^2 + c\Sigma x^3$$

$$\Sigma x^2 y = a\Sigma x^2 + b\Sigma x^3 + c\Sigma x^4$$

x	y	xy	x^2	x^3	x^2y	x^4
1	1000	1000	1	1	1000	1
2	1300	2600	4	8	5200	16
3	900	2700	9	27	8100	81
4	1100	4400	16	64	17600	256
5	800	4000	25	125	20,000	625
15	5100	14700	55	225	51900	979

Solving the equation

$a = 900$

$b = 197.142$

$c = -42.857$

$\Rightarrow y = 900 + 197.14x - 42.85x^2$

at 2025 $[x = 6]$

$y = 900 + 197.4 \times 6 - 42.85 \times 36$

$= 540.$

at 2027: $x = 8.$

$y = 900 + 197.14(8) - 42.85(64)$

$= 265.762$

Assignment 1.

select a real data set of 16 observations with 1 dependent and one 1 independent variable. import pandas numpy matplotlib. import the dataset. print dataframe. import store the data in form of dependent and independent variables. split the dataset into training and test fit the simple linear regression model. Visualise the linear regression result using scatter and predict. use sklearn preprocessing to import polynomial features, to add polynomial terms for more accuracy. Fit polynomial regression model and visualize that. predict / forecast the data for next 4. print the whole programming code and result and submit the hardcopy & softcopy.

submission date : 20/01/2025.

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1) Fit an exponential curve $y = ae^{bx}$. 14/1/25

$\log y = \log a + bx \log e$.

$Y = AX + BX$

$Y = \log y \quad B = b \log e$.

$A = \log a$

x	1	5	7	9	12
y	10	15	12	15	21

$\Sigma y = \Sigma A + B \Sigma x$.

$\Sigma xy = A \Sigma x + B \Sigma x^2$

x	y	$y' = \log y$	x^2	$\Sigma xY'$
1	10	1.	1	1.
5	15	1.176091	25	5.880455
7	12	1.079919	49	7.55391
9	15	1.176091	81	10.584819
12	21	1.322219	144	15.866628
34	73	5.753533	300	40.885804

$5.7535 = 5A + 34B$

$40.8858 = 34A + 300B$.

$A = 0.9765$

$B = 0.0256$

$A = \log a \Rightarrow a = \text{antilog}(A) = 9.462$.

$B = b \log e \Rightarrow b = \frac{0.0256}{\log e} = 0.058$

$y = 9.462 e^{0.058x}$.

UNEQUAL INTERVALS USING

$$F(x) = \frac{(x-x_1)+(x-x_2)+(x-x_3)+\ldots}{(x_0-x_1)+(x_0-x_2)+(x_0-x_3)+\ldots} \times y_0 + \frac{(x-x_0)(x-x_2)(x-x_3)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)} \times y_1$$

$$+ \ldots \frac{(x-x_0)(x-x_1)\ldots(x-x_{n-1})}{(x_n-x_0)+(x_n-x_1)+\ldots(x_n-x_{n-1})} \times y_n.$$

? use lagrange's polynomial and find $y(10)$ from the data

x	5	6	9	11
y	12	13	14	16

$$F(x) = \frac{(x-6)(x-9)(x-11)}{(5-6)(5-9)(5-11)} \times 12 + \frac{(x-5)(x-9)(x-11)}{(6-5)(6-9)(6-11)} \times 13 +$$

$$\frac{(x-5)(x-6)(x-11)}{(9-5)(9-6)(9-11)} \times 14 + \frac{(x-5)(x-6)(x-9)}{(11-5)(11-6)(11-9)} \times 16$$

$$F(10) = \frac{(10-6)(10-9)(10-11)}{-1 \times -4 \times -6} \times 12 + \frac{5 \times 1 \times -1}{1 \times 3 \times -5} \times 13 +$$

$$\frac{5 \times 4 \times -1 \times 14}{4 \times 3 \times -2} + \frac{5 \times 4 \times 1 \times 16}{6 \times 5 \times 2}$$

$$= 2 + \frac{-13}{3} + \frac{35}{3} + \frac{16}{3} = 2 + \frac{38}{3} = 14.666$$

Interpolation with equal intervals:

1) Newtons forward (for beginning of data)

$$y(x) = y_0 + u\Delta y_0 + \frac{u(u-1)}{2!}\Delta^2 y_0 + \ldots\ldots$$

where $u = \frac{x-x_0}{h}$

2) Newtons backward (for end of data)

$$y(x) = y_n + v\Delta y_n + \frac{v(v+1)}{2!}\Delta^2 y_n + \ldots\ldots$$

$$v = \frac{x-x_n}{h}$$

3) Newtons divided difference (middle)

Suppose $y = f(x)$ takes the values $y_0, y_1, y_2 \ldots y_n$ corresponding to equally spaced values $x_0, x_1, \ldots x_n$ ie $x_i - x_{i-1} = h$.

For this type of data we use newtons forward data for interpolating the beginning and newtons backward formula for interpolating end of data

of lead and zinc where t denotes Melting point of an alloy temperature in °c.
and p is the % of lead in alloy
Find the melting point of alloy containing 84% lead and 65% lead.

p :	60	70	80	90
t :	226	250	276	304

case 1 $x_0 = 90$ $\quad v = \frac{x - x_0}{h} = \frac{84 - 90}{10} = -0.6$.
$h = 10$
$x = 84$.

Newton's Backward difference table.

p	t	∇t	$\nabla^2 t$	$\nabla^3 t$
60	226			
		24.		
70	250		2	
		26.		0.
80	276		2	
		28.		
90.	304			

$y_0 = 304. \quad \nabla y_0 = 28 \quad \nabla^2 y_0 = 2$
$\nabla^3 y_0 = 0.$

$y(84) = y_0 + v \nabla y_0 + \frac{v(v+1)}{2!} \nabla^2 y_0 + \dots$

$y(84) = 304 + -0.6 \times 28 + \frac{-0.6 \times -0.4}{2!} \times 2. + 0$

$= 304 - 16.8 - 0.24$
$= 286.96.$

For 65%
$y_0 = 226. \quad \nabla y_0 = 24 \quad \nabla^2 y_0 = 2 \quad \nabla^3 t.$
$u = \frac{65 - 60}{90 - 84} = 0.5.$

$y(65) = 226 + 0.5 \times 24 + \frac{0.5 \times -0.5 \times 2.}{2}$
$= 226 + 12 - 0.25.$
$= 237.75$

23.

Newtons divided difference method.

$\Delta F(x_0) = \frac{F(x_1) - F(x_0)}{x_1 - x_0} = F[x_0, x_1] = \Delta y$

$\Delta^2 y = F[x_i, x_{i+1}, x_{i+2}]$

$y = F(x) = y_0 + (x - x_0) F(x_0) + (x - x_0)(x - x_1) \Delta^2 F(x_0) + (x - x_2)(x - x_1)(x - x_3) \Delta^3 F(x_0) + \dots$

? Find f(4) using newtons divided difference form:

x	0	1	2	5
y	2	3	12	147

x	y	$\Delta F(x)$	$\Delta^2 F(x)$	$\Delta^3 F(x)$
0	2			
1	3	1		
2	12	9	4	1
5	147.	45	9.	

$y = F(x) = y_0 + (x-x_0)\Delta F(x_0) + (x-x_0)(x-x_1)\Delta^2 F(x_0) +$

$(x-x_0)(x-x_1)(x-x_2)\Delta^3 F(x_0)$

$F(4) = 2 + (4-0)1 + (4-0)(4-1)4 + (4-0)(4-1)(4-2)1$

$= 78.$

Taylors theorem:

If F is a function defined on [a, b]. such that.

(i) $F(x), F'(x), F''(x), \ldots F^{n-1}(x)$ are continuous on [a, b]

(ii) $F^n(x)$ exist in (a, b)

Then there exist a real no. $c \in (a, b)$ such that

$$F(x) = F(a+b) = F(a) + hF'(a) + \frac{h^2}{2!}F''(a) + \ldots + \frac{h^{n-1}}{(n-1)!}F^{n-1}(x) + \frac{h^n}{n!}F$$

$x = a+b$; $h = x-a$.

when $h = x-a$ $f(x) = F(a) + \frac{x-a}{1!}F'(a) + \frac{(x-a)^2}{2!}F''(a) + \ldots$

$\frac{(x-a)^n}{n!}F^n(c)$,

when $a = 0$, Taylor series for $F(x)$ becomes Maclaurin's series

$$F(x) = F(0) + \frac{x}{1!}F'(0) + \frac{x^2}{2!}F''(0) + \ldots \frac{x^n}{n!}F^n(c)$$

? Find a polynomial approx. for e^x, here $x=2$ accurate to ± 0.0014.

Ans) we have $\frac{e^2}{7!} = 0.0014$.

$$f(x) = e^{2+x-2} = f(a) + \frac{[x-a]}{1!} \times f'(a) + \frac{[x-a]^2}{2!} f''(x) + \cdots$$

$$= e^2 + \frac{[x-2]}{1!} e^2 + \frac{[x-2]^2}{2!} e^2 + \cdots \frac{[x-2]^6}{6!} e^2 \pm 0.0014$$

? use taylor series method to find y at $x = 0.1, 0.2, 0.3$ considering terms upto 3rd degree.

given $\frac{dy}{dx} = x^2 + y^2$, $y(0) = 1$.

$x_0 = 0$

$y(0) = 1 \Rightarrow y_0 = 1 \Rightarrow x_0 = 0$

$$y(x) = f(x) \simeq y(x_0) + \frac{x-x_0}{1!} y'(x_0) + \frac{(x-x_0)^2}{2!} y''(x_0) + \frac{[x-x_0]^3}{3!} y'''(x_0)$$

$y' = x^2 + y^2$.

$y'' = 2x + 2y y'$

$y''' = 2 + 2[y y' + y'']$.

$y_0 = 1$

$y'(x_0) = y_0' = x_0^2 + y_0^2 = 0^2 + 1^2 = 1$.

$y_0'' = 2x_0 + 2y_0 y_0' = 2 \times 0 + 2 \times 1 \times 1 = 2$.

$y_0''' = 2 + 2[y_0 y_0' + y_0'']$

$= 2 + 2[1 \times 1 + 2] = 8$

$$y(x) = 1 + \frac{x}{1!} \times 1 + \frac{x^2}{2!} \times 2 + \frac{x^3}{3!} \times 8$$

$y(0.1) = 1.1113$

$y(0.2) = 1.2507$

$y(0.3) = 1.4126$

? Expand $f(x) = e^x$ as Maclauris series.

$$e^x = f(0) + \frac{x}{1!} f'(0) + \frac{x^2}{2!} f''(0) + \cdots$$

$$= 1 + x + \frac{x^2}{2} + \cdots$$

? Expand $\log[\sin x]$ in powers of $x-2$ by taylor's series

$f(x) = f[2 + (x-2)]$

$= f(2) + \frac{x-2}{1!} f'(x) + \frac{(x-2)^2}{2!} f''(2) + \cdots$

Ideas / Numerical Integration / Trapezoidal rule

4. Gauss quadrature 2 point formulae $= \int_a^b F(x)\,dx = \frac{b-a}{2}\left[\omega_1 F(x_1) + \omega_2 F(x_2)\right]$

1.	Trapezoidal Rule	$\int_a^b F(x)\,dx$	$= \frac{h}{2}\left[(y_0+y_n) + 2(y_1+y_2+\ldots y_{n-1})\right]$
2.	Simpson's 1/3 rule	$\int_a^b F(x)\,dx$	$= \frac{h}{3}\left[(y_0+y_n) + 4(y_1+y_3+y_5+\ldots) + 2(y_2+y_4+\ldots)\right]$ (odd, even)
3	Simpson's 3/8 rule	$\int_a^b F(x)\,dx$	$= \frac{3h}{8}\left[(y_0+y_n) + 2(y_3+y_6+\ldots) + 3(y_1+y_2+y_4+y_5+\ldots)\right]$ (3n, y_{4+1+y_5})
4.	Gauss's quadrature 2 point formulae.	$\int_a^b F(x)\,dx$	$= \frac{b-a}{2}\left[\omega_1 F(x_1) + \omega_2 F(x_2)\right]$ $\omega_1 = \omega_2 = 1$ $x_1 = \frac{b-a}{2}\left[\frac{1}{\sqrt{3}}\right] + \frac{b+a}{2}$ $x_2 = \frac{b-a}{2}\left[\frac{1}{\sqrt{3}}\right] + \frac{b+a}{2}$
5	Gauss Quadrature 3 point formulae	$\int_a^b F(x)\,dx$	$= \frac{b-a}{2}\left[\omega_1 F(x_1) + \omega_2 F(x_2) + \omega_3 F(x_3)\right]$ $\omega_1 = 5/9 = \omega_3$; $\omega_2 = 8/9$ $x_1 = \frac{b-a}{2} c_1 + \frac{b+a}{2}$ $\quad c_1 = -\sqrt{3/5}$ $x_2 = \frac{b-a}{2} c_2 + \frac{b+a}{2}$ $\quad c_2 = 0$ $x_3 = \frac{b-a}{2} c_3 + \frac{b+a}{2}$ $\quad c_3 = \sqrt{3/5}$
6	Initial value problem IVP $\frac{dy}{dx} = F(x,y)$. $y(x_0) = y_0$	To find $y_1, y_2, \ldots$	y_1? $\quad y_2$? (x_0, y_0) Given $\quad x_0+h = x_1 \quad x_0+2h = x_2$
6.1	Euler's method	y_{n+1}	$y_n + hF(x_n, y_n)$
6.2	Runge kutta method of 2nd order	k_1 k_2 	$= hF(x_0, y_0)$ $= hF[x_0+h, y_0+k_1]$ $\Delta y = k = \frac{1}{2}[k_1+k_2]$ $y_1 = y_0 + \Delta y$
6.3	Runge kutta method of 4th order.	k_1 k_2 k_3 k_4 k	$= hF[x_0, y_0]$ $= hF[x_0+h/2, y_0+k_1/2]$ $= hF[x_0+h/2, y_0+k_2/2]$ $= hF[x_0+h, y_0+k_3]$ $= \Delta y = \frac{1}{6}[k_1+2k_2+2k_3+k_4]$

$y_1 = y_0 + \Delta y$

1. Evaluate $\int_0^6 \frac{dx}{1+x^2}$ using

(i) Trapezoidal rule.
(ii) Simpson's 1/3 rule.
(iii) Simpson's 3/8 rule.

(i) $f(x) = y(x) = \frac{1}{1+x^2}$.

$h = \frac{6-0}{6} = 1$.

$x_0 = 0$. $x_0 + h$. h.

x	0	1	2	3	4	5	6
y	1	1/2	1/5	1/10	1/17	1/26	1/37

$y = \frac{1}{2}\left[1 + \frac{1}{37} + 2\left[\frac{1}{2} + \frac{1}{5} + \frac{1}{10} + \frac{1}{17} + \frac{1}{26}\right]\right]$

$= \frac{2.82159}{2} = 1.410795$.

(ii) $\int_0^6 f(x)\,dx = \frac{1}{3}\left[\left(1 + \frac{1}{37}\right) + 4\left[\frac{1}{2} + \frac{1}{10} + \frac{1}{26}\right] + 2\left[\frac{1}{5} + \frac{1}{17}\right]\right]$

$= 1.366$

(iii) $\int_0^6 y\,dx = \frac{3}{8}\left[1 + \frac{1}{37} + 2\left[\frac{1}{10}\right] + \underbrace{1.02710.}\right.$

$\left. 3\times\left[\frac{1}{2} + \frac{1}{5} + \frac{1}{17} + \frac{1}{26}\right]\right]$

$= 1.356707$

Actual value $= 1.41056$.

Limits of error

2. Use Gaussian Quadrature 2 point and 3 point Formulae to evaluate $\int_2^4 x^4 + 1\,dx$ and compare with actual value

$a = 2$ $b = 4$ $b - a = 2$. 2 point

$\int_2^4 y\,dx = \frac{2}{2}\left[1 \cdot F(x_1) + F(x_2)\right]$.

$= F[x_1] + F(x_2)$

$x_1 = 1 \times \left[-\frac{1}{\sqrt{3}}\right] + \frac{1}{3} = 3 - \frac{1}{\sqrt{3}}$.

$x_2 = 1 \times \frac{1}{\sqrt{3}} + 3 = 3 + \frac{1}{\sqrt{3}}$.

$\int_2^4 y\,dx = F\left[3 - \frac{1}{\sqrt{3}}\right] + F\left[3 + \frac{1}{\sqrt{3}}\right]$

$= \frac{3\sqrt{3}-1}{\sqrt{3}} \cdot \frac{3\sqrt{3}+1}{\sqrt{3}}$.

$= F(9 - \sqrt{3})(3 + \sqrt{3})$

$= 200.222$

Actual value $= 200.41$

3 point Formulae:

$1 \times \left[\frac{5}{9}F[x_1] + \frac{8}{9}F[x_2] + F[x_3]\frac{5}{9}\right]$

$x_1 = c_1 + 3 = 3 - \sqrt{\frac{3}{5}}$.

$x_2 = c_2 + 3 = 3$.

$x_3 = c_3 + 3 = 3 + \sqrt{\frac{3}{5}}$.

$= 200.41$

1.43 5.40 5.55 4.78

6 Apply Euler's method to find $y[0.2]$ from the initial value problem

$\frac{dy}{dx} = x+y+xy$, $h = 0.1$, $y(0) = 1$

$y_0 = 1$, $y_1 = ?$, $y_2 = ?$

$x_0 = 0$, 0.1, 0.2, 0.3

$y_{n+1} = y_n + hF[x_n, y_n]$

$y_1 = y_0 + hF[x_0, y_0]$ $\quad y_2 = y_1 + hF[x_1, y_1]$

$y_0 = 1$ $\quad y_1 = 1 + 0.1 \times [0+1+0] = 1+0.1 = 1.1$

$y_2 = 1.1 + 0.1 \times F[0.1, 1.1] = 1.1 + 0.1 \times [0.1 + 1.1 + 1.1 \times 0.1]$

$= 1.1 + 0.1 \times [1.2 + 0.11]$

$= 1.1 + (0.1 \times 1.31)$

$= 1.1 + 0.131$

$= 1.231$

$\left[\frac{dy}{dx} = x+y+xy = F(x,y)\right]$

---

7 Apply Rungekutta method of 2nd order and 4th order to find $y[0.2]$ from IVP $\frac{dy}{dx} = x+y$, $y(0) = 1$, $h = 0.1$

$k_1 = hF[x_0, y_0]$

$k_2 = hF[x_0+h, y_0+k_1)$ $\quad \Delta y = k = \frac{1}{2}[k_1+k_2]$ $\quad y_1 = y_0 + k$.

let $y(0) = 1$ $x_0 = 0$ $\quad$ // 2nd order //

$k_1 = 0.1 \times [1+0] = 0.1$

$k_2 = 0.1 \times [F[0.1, 1.1]] = 0.1 \times [0.1+1.1] = 0.1 \times 1.2 = 0.12$

$\Delta y = k = \frac{1}{2}[0.1+0.12] = \frac{0.22}{2} = 0.11$

$y_1 = y_0 + k = 1 + 0.11 = 1.11$

$k_1 = hF[x_1, y_1] = 0.1 \times F[0.1, 1.11] = 0.1 \times [1.21] = 0.121$

$k_2 = hF[x_1+h, y_1+0.121] = 0.1 \times [0.2 + 1.11 + 0.121]$

$= 0.1 \times [1.431]$

$= 0.1431$

$\Delta y = \frac{1}{2}[0.121 + 0.1431]$

$\frac{1}{2}[0.2641] = 0.1321 = k$

$y_2 = y_1 + k$

$y_2 = 1.11 + 0.1321 = 1.2421$

$y_2 = 1.242 = y[0.2]$

4th order

$k_1 = h[f(x_0, y_0)]$

$k_2 = hf\left[x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}\right]$

$k_3 = hf\left[x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}\right]$

$k_4 = hf[x_0 + h, y_0 + k_3]$

$\Delta y = k = \frac{1}{6}[k_1 + 2k_2 + 2k_3 + k_4]$

$k_1 = 0.1 \times [0 + 1] = 0.1$

$k_2 = 0.1 \times f[0.05, 1.05]$

$= 0.1 \times 1.1 = 0.11$

$k_3 = 0.1 \times f[0.05, 1.055]$

$= 0.1 \times [1.105]$

$= 0.1105$

$k_4 = 0.1 \times [0.1 + 1 + 0.1105]$

$= 0.1 \times [1.2105]$

$= 0.121505$

$\Delta y = k = 0.11034$

$y_1 = y_0 + k = 1.11034$

$k_1 = 0.1 \times [0.1 + 1.11034]$

$= 0.1 \times 1.21034 = 0.1210341$

$k_2 = 0.1[0.1 + 0.05 + 1.11034 + 0.060517]$

$= 0.132085$

$k_3 = 0.1 \times [0.1 + 0.05 + 1.11084 + 0.066042]$

$= 0.132076$

$k_4 = 0.1 \times [0.2 + 1.11034 + 0.01320]$

$= 0.144234$

$\Delta y = k = \frac{1}{6}[$

$= 0.132265$

$y_2 = y_1 + k_1 = 1.11034 + 0.13226 = 1.24260$

1). Evaluate $\int_0^{10} \frac{1}{1+x}\,dx$ using 21/1/25.

(i) Trapezoidal rule

(ii) Simpson's 1/3 rule.

(iii) Simpson's 3/8 rule.

(iv) Gauss's quadrature 2 point formula

(v) Gauss's quadrature 3 point formula

$\int_0^{10} \frac{1}{1+x}\,dx$ Take $n = 10$ step

(i) Trapezoidal rule

$= \frac{h}{2}[(y_0 + y_n) + 2[y_1 + y_2 + \dots y_{n-1}]$

$h = \frac{10-0}{10} = 1$

$x_0 = 0$

0	1	2	3	4	5	6	7	8	9	10
1	1/2	1/3	1/4	1/5	1/6	1/7	1/8	1/9	1/10	1/11

$y = \frac{1}{2}\left[1 + \frac{1}{11} + 2\left[\frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8} + \frac{1}{9} + \frac{1}{10}\right]\right]$

$= 2.474442$

(ii) Simpson's 1/3 rule

$\int_a^b f(x)\,dx = \frac{h}{3}[y_0 + y_0 + 4[y_1 + y_3 \dots] + 2[y_2 + y_4 + y_6 \dots]$

$y = \frac{1}{3}\left[1 + \frac{1}{11} + 4\left[\frac{1}{2} + \frac{1}{4} + \frac{1}{6} + \frac{1}{8} + \frac{1}{10}\right] + 2\left[\frac{1}{3} + \frac{1}{5} + \frac{1}{7} + \frac{1}{9}\right]\right]$

(iii) Simpson's 3/8 rule:

$$\int_a^b F(x)\,dx = \frac{3h}{8}\left[y_0 + y_{10} + 2\left[y_3 + y_6 + y_9\right] + 3\left[y_1 + y_2 + y_4 + y_5 + y_7 + y_8\right]\right]$$

$$= \frac{3}{8}\left[1 + \frac{1}{11} + 2\left[\frac{1}{4} + \frac{1}{7} + \frac{1}{10}\right] + \frac{3}{2} + \frac{3}{3} + \frac{3}{5} + \frac{3}{6} + \frac{3}{8} + \frac{3}{9}\right]$$

$$= 2.394358$$

(iv) Gauss's Quadrature 2 point rule

$$\int_a^b F(x)\,dx = \frac{b-a}{2}\left[w_1 F(x_1) + w_2 F(x_2)\right]$$

$w_1 = w_2 = 1$

$x_1 = \frac{b+a}{2} - \frac{(b-a)}{2}\frac{1}{\sqrt{3}}$

$x_2 = \frac{b+a}{2} + \frac{b-a}{2}\frac{1}{\sqrt{3}}$

$\frac{b-a}{2} = 5$ $\quad x_1 = 5 - 5/\sqrt{3} = 2.113248$

$\frac{b+a}{2} = 5$ $\quad x_2 = 5 + 5/\sqrt{3} = 7.886751$

$F[x_1] = 0.3212079$

$F[x_2] = 0.1125270$

$y = 5 \times [0.3212079 + 0.1125270]$

$= 2.16882.$

(v) Gauss 3 point formula:

$$\int_a^b F(x)\,dx = \frac{b-a}{2}\left[w_1 F(x_1) + w_2 F(x_2) + w_3 F(x_3)\right]$$

$w_1 = w_3 = 5/9 \quad w_2 = 8/9$

$x_1 = \left(5 \times -\sqrt{\frac{3}{5}}\right) + 5 = 1.127016$

$x_2 = 5$

$x_3 = 5\sqrt{\frac{3}{5}} + 5 = 8.872983$

$y = 5 \times \left[\frac{5}{9} F(x_1) + \frac{8}{9} F(x_2) + \frac{5}{9} F(x_3)\right]$

$F(x_1) = 0.471014$

$F(x_2) = 0.16666$

$F(x_3) = 0.101286.$

$y = 5 \times \left[\frac{5}{9} \times 0.471014 + \frac{8}{9} \times 0.16666 + \frac{5}{9} \times 0.101286\right]$

$= 2.32800$

? Find $y[0.4]$ from the ODE $\frac{dy}{dx} = x + y + xy$

$h = 0.2$ $\quad y(0) = 1.$

using (1) Euler's method

(2) Rungekutta method of 2nd order.

(3) RK method of 4th order.

$\frac{dy}{dx} = x + y + xy$

$y(0) = 1$

$y_0 = 1 \quad y_1 = ? \quad y_2 = ? \quad y_3 = ? \quad y_4 = ?$

$x_0 = 0 \quad 0.2 \quad 0.4 \quad x_3 \quad x_4$

$y_{n+1} = y_n + h F[x_n, y_n]$

$y_1 = y_0 + h \times F[x_0, y_0]$

$y_1 = 1 + 0.2[0 + 1 + 0]$

$= 1.2$

$y_2 = y_1 + h \times F[0.2, 1.2]$

$= 1.2 + 0.2[0.2 + 1.2 + 0.24]$

$= 1.528.$

$y[0.4] = 1.528$

2

$k_1 = h \times F[x_0, y_0]$

$k_2 = h \times F[x_0 + h, y_0 + k_1]$

$\Delta y = k = \frac{k_1 + k_2}{2}$ $\quad y_0 = 1$

$y_2 = y_1 + k'$

$y_1 = y_0 + k$

$k_1 = 0.2 \times [0 + 1 + 0] = 0.2$

$k_2 = 0.2 \times F[0.2, 1.2]$

$= 0.2 \times [1.4 + 0.24]$

$= 0.2 \times [1.64] = 0.328.$

$k = \frac{0.2 + 0.328}{2} = 0.264$

$y_1 = 1 + 0.264 = 1.264.$

y_2 :

$k_1 = 0.2 \times F[0.2, 1.264]$

$= 0.2 \times [0.2 + 1.264 + 0.2528]$

$= 0.84336.$

$k_2 = 0.2 \times F[0.4, 1.60736]$

$= 0.2 [2.650304] = 0.530060.$

$k = 0.4367104$

$y_2 = y_1 + k.$

$= 1.264 + 0.43671$

$= 1.7.$

(3)

$k_1 = h \times F[x_0, y_0].$

$k_2 = h \times F[x_0 + h/2, y_0 + k_1/2]$

$k_3 = h \times F[x_0 + h/2, y_0 + k_2/2].$

$k_4 = h \times F[x_0 + h, y_0 + k_3]$

$k = \Delta y = \frac{1}{6}[k_1 + 2k_2 + 2k_3 + k_4]$

$k_1 = 0.2[1] = 0.2.$

$k_2 = 0.2 \times F[0.1, 1.1].$

$= 0.2[0.1 + 1.1 + 0.11]$

$= 0.262.$

$k_3 = 0.2 \times F[0.1, 1.131]$

$= 0.26882$

$k_4 = 0.2 \times F[0.2, 1.26882].$

$= 0.34451.$

$\Delta y = \frac{1}{6}. 0.2677691$

$y_1 = y_0 + k = 1.2677691.$

$k_1 = 0.2 \times F[0.2, 1.2677691]$

$= 0.34424.$

$k_2 = 0.2 \times F[0.3, 1.439811]$

$= 0.43435.$

$k_3 = 0.2 \times F[0.3, 1.4841866]$

$= 0.44606.$

$k_4 = 0.2 \times F[0.4, 1.7113751]$

$= 0.55985$

$\Delta y = 0.4441511$

$y_2 = y_1 + k = 1.711$

? Derive Gauss quadrature 2 point formula : 22/1/25

let $\int_a^b F(x)dx = c_1 F[x_1] + c_2 F[x_2]$ Here there are 4 unknown c_1, x_1, c_2, x_2. so we take $F[x] = a_0 + a_1 x + a_2 x^2 + a_3 x^3$.

so RHS $= c_1 F(x_1) + c_2 F(x_2) = c_1 [a_0 + a_1 x_1 + a_2 x_1^2 + a_3 x_1^3] + c_2 [a_0 + a_1 x_2 + a_2 x_2^2 + a_3 x_2^3]$ ——(1)

LHS $= \int_a^b a_0 + a_1 x + a_2 x^2 + a_3 x^3 dx$

$= a_0[b-a] + a_1\left[\frac{b^2-a^2}{2}\right] + a_2\left[\frac{b^3-a^3}{3}\right] + a_3\left[\frac{b^4-a^4}{4}\right]$ ——(2)

comparing —(1) and (2)

$c_1 + c_2 = b - a$ [coefficient of a_0]

$c_1 x_1 + c_2 x_2 = \frac{b^2-a^2}{2}$ $\quad c_1 x_1^2 + c_2 x_2^2 = \frac{b^3-a^3}{3}$ $\quad c_1 x_1^3 + c_2 x_2^3 = \frac{b^4-a^4}{4}$

By using elimination method, we can solve this 4 equations and we get

$c_1 = c_2 = \frac{b-a}{2}$ $\quad x_1 = \frac{b-a}{2}\left[\frac{-1}{\sqrt{3}}\right] + \frac{b+a}{2}$

$x_2 = \frac{b-a}{2}\left[\frac{1}{\sqrt{3}}\right] + \frac{b+a}{2}$

NUMERICAL SOLUTION OF BOUNDARY VALUE PROBLEM [BVP] USING FINITE DIFFERENCE METHOD

23/1/25.

$y(x_0) = b$, $y(x_n) = b_2$ is given. The BVP is to find y values at intermediate points of x. The BVP DE can converted to difference equation with a substitution $y' = \frac{y_{i+1} - y_{i-1}}{2h}$.

$y'' = \frac{y_{i-1} - 2y_i + y_{i+1}}{h^2}$

$i = 1, 2, 3, \ldots n-1$

? solve $y'' - y = x$, $y(0) = y(1) = 0$ using finite difference method. dividing the interval to 4 equal parts.

$y_0 = 0$ $\quad$ $y_4 = 0$

$x_0 = 0$ $\quad$ x_1 1/4 $\quad$ x_2 1/2 $\quad$ x_3 3/4 $\quad$ $x_4 = 1$

Given $n = 4$

$h = \frac{1-0}{4} = \frac{1}{4}$.

$y_i'' - y_i = x_i$

$\frac{y_{i-1} - 2y_i + y_{i+1}}{h^2} - y_i = x_i$ $\quad [i = 1, 2, 3]$

$\left[y_{i-1} - 2y_i + y_{i+1} - \frac{1}{16} y_i\right] = \frac{1}{16} x_i$

when $i = 2$

$y_1 - 2y_2 + y_3 - \frac{1}{16} y_2 = \frac{2}{32}$.

$y_0 - 2y_1 + y_2 - \frac{1}{16} y_1 = \frac{x_1}{16}$.

$y_1 - y_2 \left[\frac{33}{16}\right] + y_3 = \frac{1}{32}$ — (2)

$-y_1 \left[\frac{33}{16}\right] + y_2 = \frac{x_1}{16} = \frac{1}{64}$ — (1)

when $i = 3$

$y_2 - 2y_3 + y_4 - \frac{1}{16} y_3 = \frac{3}{64}$

$y_2 - \frac{33}{16} y_3 = \frac{3}{64}$ — (3)

$y_1 =$ $\quad$ $y_2 =$ $\quad$ $y_3 =$

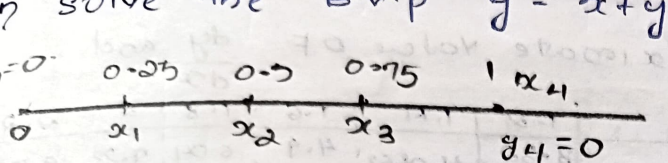
-0.0349 $\quad$ -0.0563 $\quad$ -0.0500.

-0.0349

$y_2 - \frac{33}{16} y_3 = \frac{3}{64}$ — (3)

---

7 solve the B.V.P $y' = x+y$ $y(0) = y(1) = 0$ $h = 0.25$.



$y' = x+y$. $\quad y' = \frac{y_{i+1} - y_{i-1}}{2h}$.

$\frac{y_{i+1} - y_{i-1}}{2h} = x_i + y_i$; $i = 1, 2, 3$

$i = 1$ $\quad \frac{y_2 - y_0}{0.5} = x_1 + y_1 = 0.25 \times 0.5$;

$i = 2$

$y_3 - y_1 = (x_2 + y_2) 0.5$.

$y_3 - y_1 = 0.25 + 0.5 y_2$.

$i = 3$.

$y_4 - y_2 = 0.5 [x_3 + y_3]$

$y_2 - y_0 = 0.25 \times 0.5 + 0.5 y_1 \Rightarrow$

$y_2 - 0.5 y_1 = 0.125$ — (1)

$y_3 - y_1 - 0.5 y_2 = 0.25$ — (2).

$-y_2 - 0.5 y_3 = 0.375$ — (3)

$y_1 = -0.5833$

$y_2 = -0.1666$

$y_3 = -0.4166$

Numerical Differentiation

1. Newton's forward formula.

$$\left[\frac{dy}{dx}\right]_{x=x_0} = \frac{1}{h}\left[\Delta y_0 - \frac{\Delta^2 y_0}{2} + \frac{\Delta^3 y_0}{3} - \frac{\Delta^4 y_0}{4} + \cdots\right]$$

(starting)

$$\left[\frac{d^2y}{dx^2}\right]_{x=x_0} = \frac{1}{h^2}\left[\Delta^2 y_0 - \Delta^3 y_0 + \frac{11}{12}\Delta^4 y_0 - \frac{5}{6}\Delta^5 y_0 + \cdots\right]$$

$$\left[\frac{d^3y}{dx^3}\right]_{x=x_0} = \frac{1}{h^3}\left[\Delta^3 y_0 - \frac{3}{2}\Delta^4 y_0 + \cdots\right]$$

2. Newton's backward form.

$$\left[\frac{dy}{dx}\right]_{x=x_n \text{ ending pt}} = \frac{1}{h}\left[\nabla y_n + \frac{\nabla^2 y_n}{2} + \frac{\nabla^3 y_n}{3} + \frac{\nabla^4 y_n}{4} + \cdots\right]$$

$$\left[\frac{d^2y}{dx^2}\right]_{x=x_n} = \frac{1}{h^2}\left[\nabla^2 y_n + \nabla^3 y_n + \frac{11}{12}\nabla^4 y_n + \frac{5}{6}\nabla^5 y_n + \cdots\right]$$

$$\left[\frac{d^3y}{dx^3}\right]_{x=x_n} = \frac{1}{h^3}\left[\nabla^3 y_n + 3/2\,\nabla^4 y_n + \cdots\right]$$

3. Newton's central difference formula:

$$\left[\frac{dy}{dx}\right]_{x=x_0}^{\text{central pt}} = \frac{1}{h}\left[\mu\delta y_0 - \frac{1}{6}\mu\delta^3 y_0 + \frac{1}{30}\mu\delta^5 y_0 - \cdots\right]$$

μ = average of

$$\left[\frac{d^2y}{dx^2}\right]_{x=x_0}^{\text{central pt}} = \frac{1}{h^2}\left[\delta^2 y_0 - \frac{1}{12}\delta^4 y_0 + \frac{1}{90}\delta^6 y_0 \cdots\right]$$

* For the following data find the approximate value of $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ at $x=1$, $x=2.2$, $x=1.6$

x	1	1.2	1.4	1.6	1.8	2	2.2
y	2.7183	3.32 01	4.0552	4.9 530	6.04 96	7.38 91	9.025

x	y	$\Delta y = y_{n+1} - y_n$	$\Delta^2 y_n = \Delta y_{n+1} - \Delta y_n$	$\Delta^3 y_n$	$\Delta^4 y_n$	$\Delta^5 y_n$	$\Delta^6 y_n$
1	2.7183						
		0.6018					
1.2	3.3201		0.1333				
		0.7351		0.0294			
1.4	4.0552		0.1627		0.0067		
		0.8978		0.0361		0.0013	
		1/2	0.1988	1/2	0.008	1/2	0.0001
		1.0966		0.0441		0.0014	
1.8	6.0496		0.2429		0.0094		
		1.3395		0.0535			
2	7.3891		0.2964				
		1.6359					
2.2	9.0250						

$$\Rightarrow \left.\frac{dy}{dx}\right|_{x=1.} = \frac{1}{0.2}\left[0.6018 - \frac{0.1333}{2} + \frac{0.0294}{3} - \frac{0.0067}{4} + \frac{0.0013}{5} + \frac{0.0001}{6}\right]$$

$$= 2.7175$$

$$\Rightarrow \left.\frac{d^2y}{dx^2}\right|_{x=1} = \frac{1}{0.04}\left[0.1333 - 0.0294 + \frac{11}{12}(0.0067) - \frac{5}{6}(0.0013) + \frac{137}{180}\times 0.0001\right]$$

$$= 2.712$$

$$\left.\frac{dy}{dx}\right|_{x=1.6} = \frac{1}{0.2}\left[\frac{0.8978 + 1.0966}{2} - \frac{1}{6}\left[\frac{0.0361 + 0.0141}{2}\right] + \frac{1}{30}\left[\frac{0.0013 + 0.0014}{2}\right]\right]$$

$$= 4.0952$$

$$\left.\frac{d^2y}{dx^2}\right|_{x=} = \frac{1}{(0.2)^2}\left[0.2964 + \frac{0.0535}{2} + \frac{11}{12}(0.0094) + \frac{5}{6}\times 0.0014 + \frac{137}{180}\times 0.0001\right]$$

$$= 8.99$$

$$\left.\frac{dy}{dx}\right|_{x=2.2.} = \frac{1}{0.2}\left[1.6359 + \frac{0.2964}{2} + \frac{0.0535}{3} + \frac{0.0094}{4} + \frac{0.0014}{5} + \frac{0.0001}{6}\right]$$

$$= 9.022$$

$$\left.\frac{d^2y}{dx^2}\right|_{x=1.6} = \frac{1}{0.04}\left[0.1988 - \frac{1}{12}(0.0099) + \frac{1}{90}\times 0.0001\right]$$

$$= 4.9583$$

Open book exam Module 2

1) P.T $\Delta \log F(x) = \log\left[1 + \frac{\Delta F(x)}{F(x)}\right]$ $\left(\log\left[\frac{\Delta F(x) + F(x)}{F(x)}\right]\right)$

$\Delta \log F(x) = \log F(x+h) - \log F(x)$

$= \log\left[\frac{F(x+h)}{F(x)}\right]$ — (1)

$\Delta F(x) = F(x+h) - F(x)$

(1) $\Rightarrow$ $\Delta \log F(x) = \log\left[\frac{\Delta F(x) + F(x)}{F(x)}\right]$

$\Delta \log F(x) = \log\left[1 + \frac{\Delta F(x)}{F(x)}\right]$

---

2. Evaluate $\int_0^1 \frac{dx}{1+x^2}$ using simpsons 1/3 rule: $h = 1/4$

x	$x_0 = 0$	$x_1 = 0.25$	$x_2 = 0.5$	$x_3 = 0.75$	$x_4 = 1$
$y = \frac{1}{1+x^2}$	$y_0 = 1$	$y_1 = \frac{1}{1+(0.25)^2} = 0.9411$	$\frac{1}{1+0.5^2} = 0.8$	$\frac{1}{0.75^2+1} = 0.64$	$\frac{1}{1+1^2} = 0.5$

$\int_a^b F(x)\,dx = \frac{h}{3}\left[(y_0+y_n) + 4(y_1+y_3) + 2(y_2)\right]$

$= \frac{1}{4\times 3}\left[(1+0.5) + (0.9411+0.64)\times 4 + 2[0.8]\right]$

$= 0.7853$

---

3. The velocity v (km/min) of a moped which starts from rest is given at fixed intervals of time t (min) as follows

t	2	4	6	8	10	12	14	16	18	20
v	10	18	25	29	32	20	11	5	2	0

Estimate distance covered in 20 min

$v = \frac{ds}{dt} \Rightarrow ds = v\,dt$ $\quad s = \int_0^{20} v\,dt$

Here $n = 10$.

$h = \frac{20-0}{10} = 2$

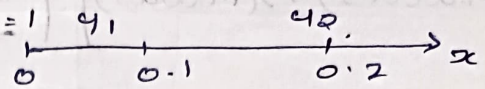
$\int_0^{20} v\,dt = \frac{2}{2}\left[10+0\right] + 2\left[18+25+29+32+20+11+5+2\right]$

(Trapezoidal rule)

$= 294$ km

4 Solve $y' = y - \frac{2x}{y}$, $y(0)=1$ in the range $0 \le x \le 0.2$.

Using euler's method, take $h = 0.1$



$y(0)=1 \quad x_0 = 0$

$y_{n+1} = y_n + hF[x_n, y_n]$

$y_1 = y_0 + hF[x_0, y_0]$

$= 1 + 0.1\left(1 - \frac{2\times 0}{1}\right)$

$= 1.1$

$y_2 = y_1 + hF(x_1, y_1)$

$= 1.1 + 0.1\left[1.1 - \frac{2\times 0.1}{1.1}\right] = 1.19181$

$x_2 = 0.2$

---

5. Apply Gauss-Seidal iteration method to solve

$20x + y - 2z = 17 \quad 3x + 20y - z = -18 \quad 2x - 3y + 20z = 25$

$$\begin{bmatrix} 20 & 1 & -2 \\ 3 & 20 & -1 \\ 2 & -3 & 20 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 17 \\ -18 \\ 25 \end{bmatrix}$$

$x^{(n+1)} = \frac{1}{a_{11}}[b_1 - a_{12}y - a_{13}z] = \frac{1}{20}[17 - y + 2z]$

$y^{(n+1)} = \frac{1}{20}[-18 - 3x + z]$

$z^{(n+1)} = \frac{1}{20}[25 - 2x + 3y]$

* Assume $x^{(0)} = y^{(0)} = z^{(0)} = 0$

$x^{(1)} = \frac{17}{20} = 0.85$

$y^{(1)} = \frac{1}{20}[-18 - 3(0.85)]$

$= -1.0275$

$z^{(1)} = \frac{1}{20}[25 - 2(0.85) + 3(-1.0275)]$

$= 1.0108$

$x^{(2)} =$

$\frac{1}{20}[17 + 1.0275 + 2(1.0108)]$

$= 1.00024$

$y^{(2)} =$

$\frac{[-18 - 3(1.00024) + 1.0108]}{20}$

$= -0.999821$

$z^{(2)} =$

$\frac{1}{20}[25 - 2[1.00024] + 3(-0.9982)]$

$= 0.9997$

$x^{(3)} =$

$\frac{1}{20}[17 + 0.9998 + 2(0.9997)]$

$= 0.99996$

$y^{(3)} =$

$\frac{[-18 - 3(0.99996) + 0.9997]}{20}$

$= -1.000005$

$z^{(3)} =$

$\frac{1}{20}[25 - 2(0.99996) + 3(-1.000005)]$

$= 1.000003$

$x^{(4)}$

$\frac{1}{20}\left[17-(-1.000005)+2(1.000003)\right]$

$= 1.0000005$

$y(4) =$

$\left(\frac{-18-3(1.0000005)+1.000003}{20}\right)$

$= -0.9999$

$x^{(4)}$

$[25-2(1.0000005)+3(-0.9999)] \times \frac{1}{20}$

$= 1.00001$

---

[6] use Newton's divided difference formula to find $F(7)$ if $F(9) = 24$ $F(5) = 120$, $F(8) = 504$, $F(9) = 720$ $F(12) = 1716$

x	3	5	8	9	12
y	24	120	504	720	1716

x	y	$\Delta F(x) = \delta[x_i, x_{i+1}]$	$\Delta^2 F(x)$	$\Delta^3 F(x)$	$\Delta^4 F(x)$
3	24				
		48			
5	120		16		
		128		1	
8	504		22		0
		216		1	
9	720		29		
		332			
12	1716				

$y = F(x) = y_0 + [x-x_0]\,\Delta F(x) + (x-x_0)(x-x_1)\,\Delta^2 F(x_0) + (x-x_0)(x-x_1)(x-x_2)\,\Delta^3 F(x_0)$

$F(7) = 24 + (7-3)\,48 + (7-3)(7-5)\,16 + (7-3)(7-5)(7-8)\,1$

$= 336$

---

4. using lagrange's formula for interpolation find $f(4)$

x	0	2	3	6
F(x)	-4	2	14	158

$F(x) = \frac{(x-x_1)(x-x_2)(x-x_3)}{x_0-x_1(x_0-x_2)(x_0-x_3)}\,y_0 + \frac{(x-x_0)(x-x_2)(x-x_3)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)} \times y_1 +$

$\frac{(x-x_0)(x-x_1)(x-x_3)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)}\,y_2 + \frac{(x_4-x_0)(x-x_1)(x-x_2)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)}\,y_3$

$= \frac{4\times2\times1}{3\times2\times3}(-4) + \frac{4\times2\times1}{3\times2\times3} = \frac{-2\times2\times1\times-4}{-2\times-3\times-6} + \frac{4\times1\times-2}{1\times-1\times-4} \times 2$

$+ \frac{4\times2\times(-2)}{3\times1\times-3}\,14 + \frac{4\times2\times1}{6\times4\times3} \times 158$

$= 89.9999$